

Introduction

This is the report to Assignment 2 of Data Science for Business II, in which the tasks are related to transform semi-raw data into time-to-event data. In addition to the data transformation, the survival analysis is also conducted with the help of 'lifelines', 'scikit-survival' and 'eli5' packages in python. This report contains survival analysis specification and Interpretations to the data and the results.

Analysis specification

The inclusion criteria state the event that patients should satisfy to be included in the study. In the case of the assignment, the event contains all those that passed away. All the patients could have been followed throughout the time interval for the study. **The outcome** includes all the events are contains all the possible description to death. **The time origin** refers to the time when the "clock started". **The entry time** specifies the time of entry of every patient. As the confirmation of covid is random among the population, both the time origin and entry time are the time that the patients are confirmed positive. **The survival time** are determined by the specification of inclusion criteria, outcome, time origin and censoring rules for each patient. It is the time from the confirmation to the death in this assignment. In this assignment, **censoring rule** refers to the case if patients are withdrawn at the end of study or if the study ends before the event occurs. **The period observation** refers to the time from presentation to when the events are occurred or censored. In this case, **the period observation** is the period between the patients are confirmed covid positive to their death or the download day.

Data Exploration

The data are part of the pre-processed semi-raw data on Covid-19 outbreak. The data were downloaded on February 22, 2021, by the teaching team.

Before conducting survival analysis, the semi-raw data are converted into time-to-event data. The new dataset contains "age", "sex", "chronic_disease_binary", "event" and "time". The exploratory data analysis conducted for this converted dataset are shown in the table 1.

There are 572405 records in this time-to-event dataset, among which there are 1368 (before deleting those with no time recorded) or 1327 (after deleting no value items) records are confirmed as passed-away, they are recorded under either "death", "died", "dead" or "deceased". Thus, the death rate for this clean time-to-event dataset is 0.23. The mean age for the data is 45.7 years old (rounded to 1 digit), and the average time period is 282 days.

Table 1 Exploratory data analysis

	age	event	time
count	572405.000000	572405.000000	572405.000000
mean	45.688383	0.002318	282.065216
std	19.708668	0.048093	73.408390
min	0	0	0
25%	29.000000	0.000000	269.000000
50%	47.000000	0.000000	293.000000
75%	59.000000	0.000000	330.000000
max	121.000000	1.000000	417.000000

To see more clearly the data, figure 1 illustrated the non-Boolean variables before transferring the Boolean variable into binary variables.

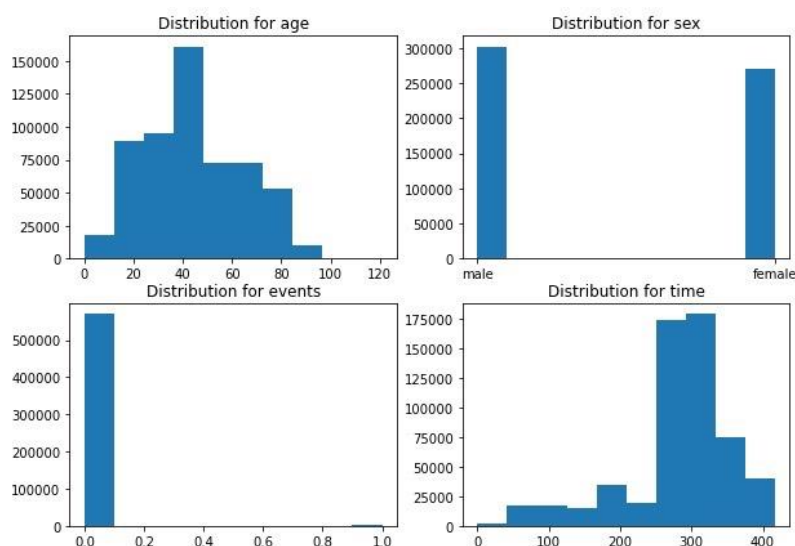


Figure 1 Distributions for variables

The distribution for age is positively skewed, while the distribution for time are negatively skewed. The dataset contains more males than females, but the discrepancy on numbers are not significantly large.

Survival analysis

Survival analysis with Kaplan-Meier (KM) estimator are carried out in this assignment with the python packages called "lifelines", "scikit-survival" and "eli5".

Kaplan-Meier estimator

The figure 2 shows the Kaplan-Meier estimator. The y-axis represents the probability a patient is still alive after time t on the x-axis. The KM curve declines as the time increases, which means that as the time

increase, the patients are less likely to survive. The flat part between $t=350$ to about $t=370$ shows that in this time of period, there are events censored instead of occurred.

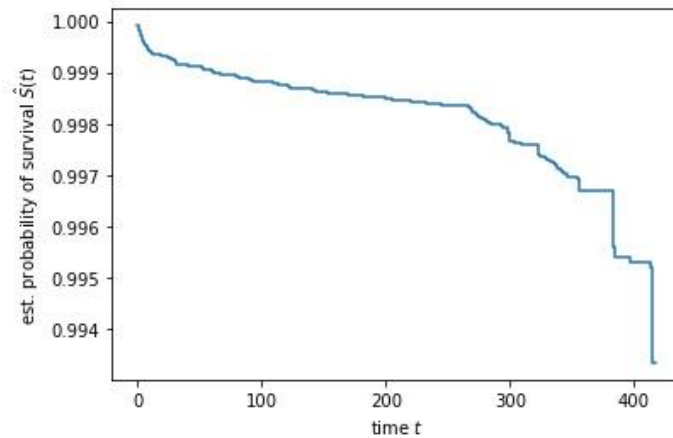


Figure 2 Kaplan-Meier estimator

The survival analysis are also analyzed on the gender and on age group base.

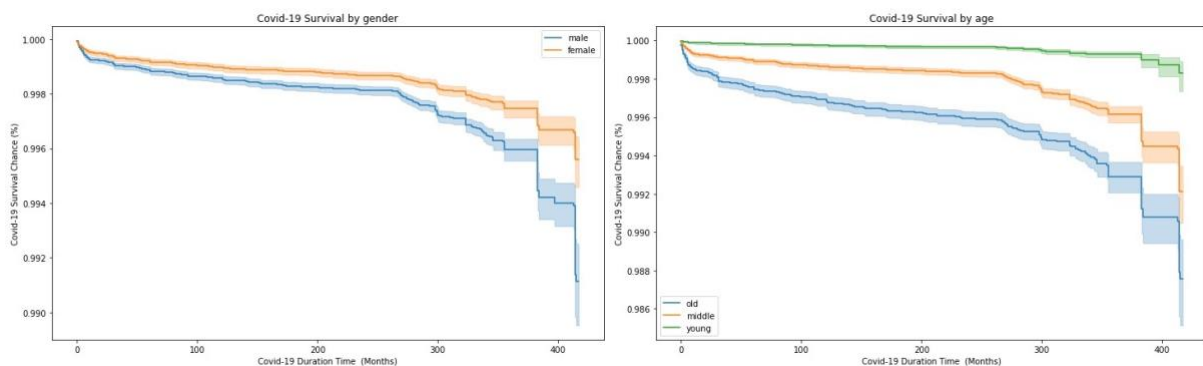


Figure 3 Covid-19 survival by gender and by age

In both the curve by gender and by age, the y-axis represents the probability a patient is still alive after time t on the x-axis. The curves decline as the time increase. In the figure by gender, the female is more likely to survival then male at the same duration time. In the figure by age, at the same duration time, the younger the patient, the higher the estimated survival probability.

The log-rank test and pair log-rank test are done respectively to these two cases. In the log-rank test, the p-value from the log-rank test equals to $8.850868e-15$, which is much less than 0.001. This indicates that the difference in survival between gender groups was statistically highly significant.

Table 2 Logrank test for KM curve on gender

test_statistic	P	$-\log_2(p)$
60.136348	$8.850868e-15$	46.683102

In the pairwise log-rank test, the test statistic is quite large and all the p-value from the pairwise log-rank test are much less than 0.001. This indicates that the difference in survival between age groups was statistically highly significant.

Table 3 pairwise logrank test

		test_statistic	P	-log2(p)
middle	old	135.380119	$2.726418e - 31$	101.532764
	young	312.978261	$4.902758e - 70$	230.241373
old	young	777.443860	$4.327723e - 171$	565.936096

For both of these two cases, the null hypothesis should be rejected, which means that survival rate for different genders could differ.

Cox model

The Cox models are plotted for two situations. The left graph in figure 4 shows the model with "sex" as the only covariate, while in the right graph, "sex" are treated as a stratify variable and "age groups" is the covariate.

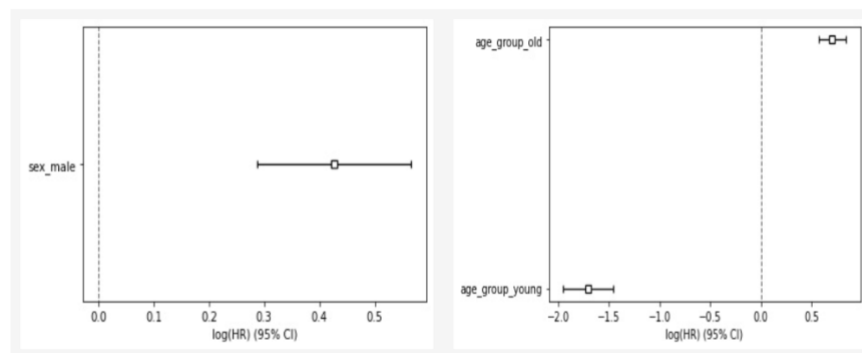


Figure 4 Cox model illustration

The coefficient for sex_male, age_group_old and age_group_young are 0.43, 0.7 and -1.71 respectively in the 2 situations. The sex "male" and "old" in age group is positively related to risk of death, while "young" is negatively related to it. However, the p-values (table 4) for them are all less than 0.005, which indicates that the difference in survival probabilities is statistically highly significant. The null hypotheses in both cases should be rejected. The survival rate for different genders and age groups could differ.

Table 4 summary for Cox models

	covariate	coef	exp(coef)	se(coef)	z	p	-log2(p)
sex as a covariate	sex_male	0.43	1.53	0.07	6.03	< 0.005	29.23
sex as a stratify	age_group_old	0.70	2.02	0.07	10.13	< 0.005	77.64
	age_group_young	-1.71	0.18	0.13	-13.53	< 0.005	136.08

Hazard assumption check

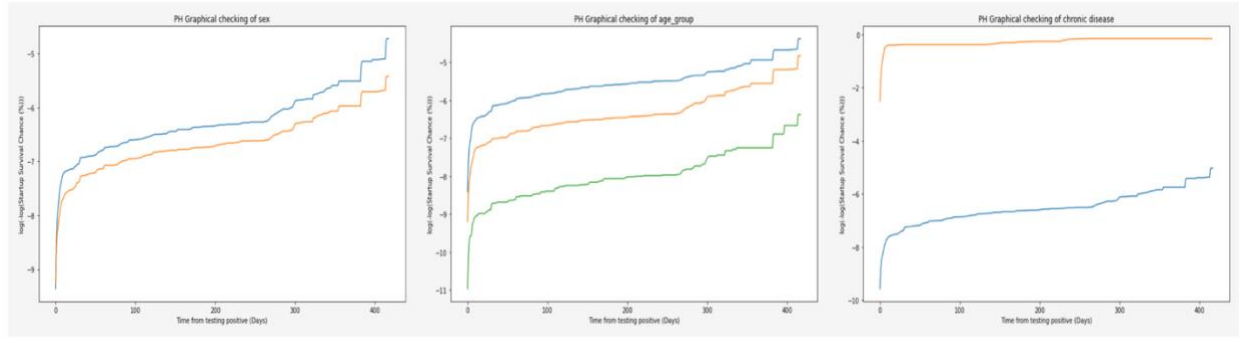


Figure 5 assumption check

The graphs for sex, age group and chronic disease are all parallel. This means for all these 3 assumptions, the assumptions hold.

Random survival forest

The random survival forest was constructed with 100 estimators and 3 iterations with scikit-survival package in python. The score with test data for the forest is approximately 0.73. The weights for the important features are shown in table 5 below.

Table 5 Weights for important features

Weight	<i>Feature</i>
0.0704 ± 0.0122	age_group_young
0.0701 ± 0.0074	age_group_old
0.0178 ± 0.0000	chronic_disease_True
0.0113 ± 0.0084	sex_male

Conclusion

The KM curves shows the estimated survival probabilities for different assumptions. It provided that female is more likely to survive and the younger the age, the higher the survival probabilities at the same duration time. The Cox models examined the survival probabilities on different covariate and found that the risk of death increase if the gender is male and if the age is older. However, the p-values for all tests provided the results are statistically highly significant, and the hypothesis should be rejected.